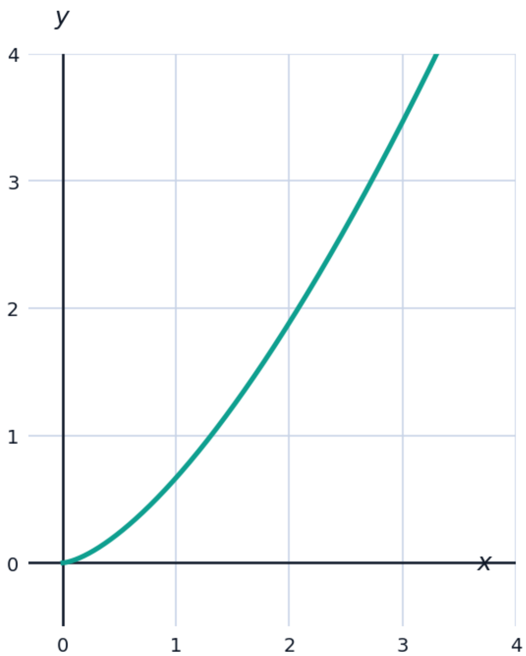
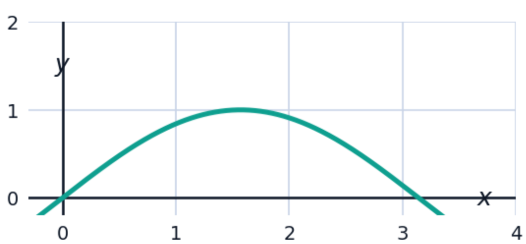
Arc Length and Improper Integrals

Arc length

1 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.



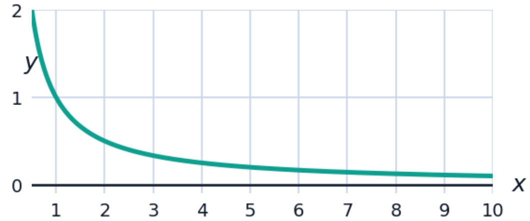
2 Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.



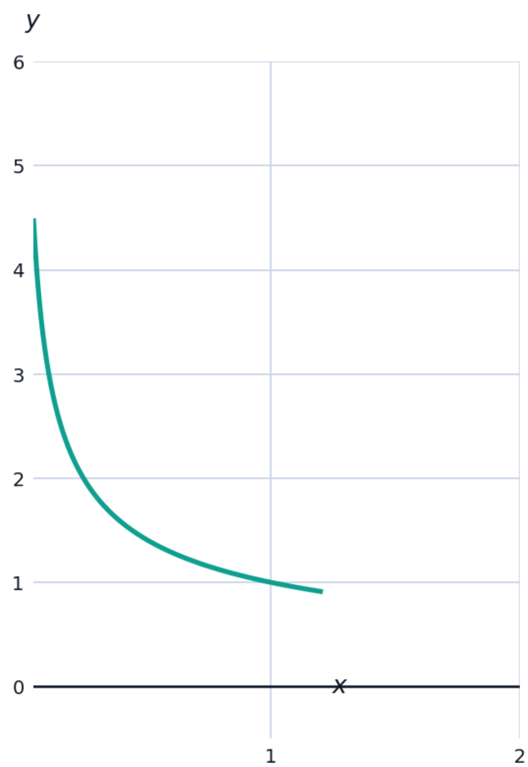
Improper integrals

3 Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$, or state that it diverges.

4 Evaluate $\int_1^{\infty} \frac{1}{x}\,dx$, or state that it diverges.

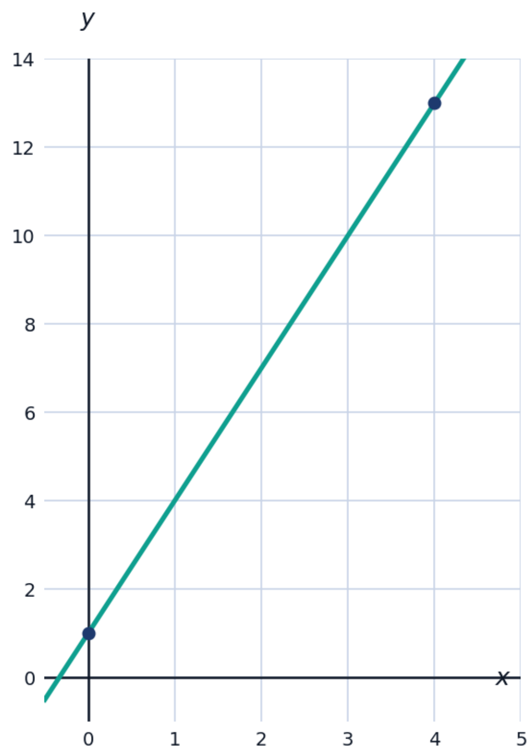


5 Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$ (improper at the lower limit).

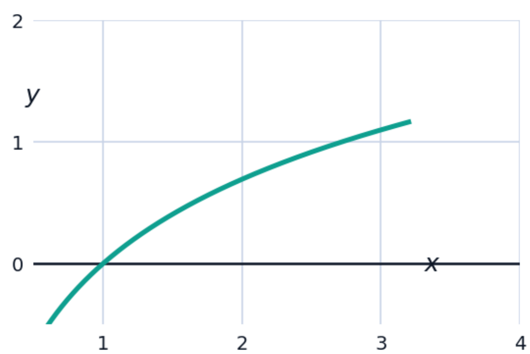


More arc length practice

6 Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.



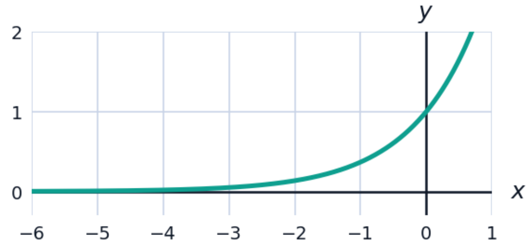
7 Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.



More improper integrals

8 Evaluate $\int_1^\infty \frac{1}{x^{1.5}}\,dx$, or state that it diverges.

9 Evaluate $\int_{-\infty}^{0} e^x \, dx$, or state that it diverges.



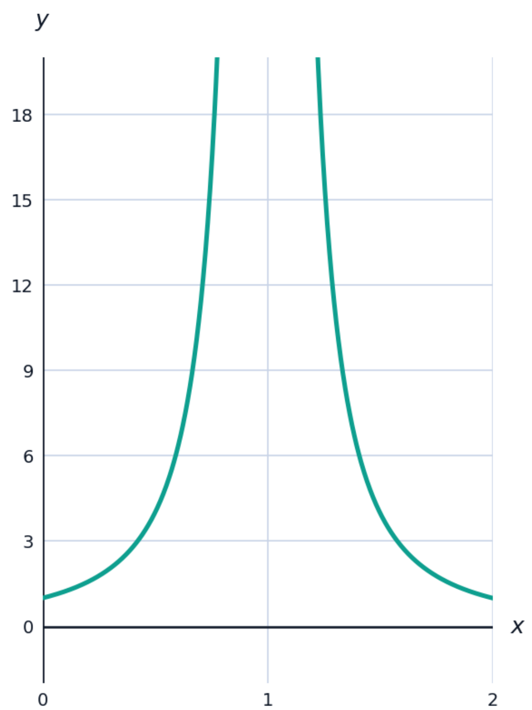
The p-integral test

10 For $\int_1^{\infty} \frac{1}{x^p} \, dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).

11 Use the p-integral result to quickly determine whether $\int_1^{\infty} \frac{1}{x^{0.9}} \, dx$ converges.

Improper integrals with infinite discontinuities inside the interval

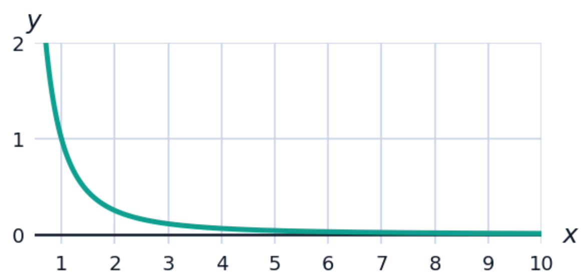
12 Evaluate $\int_0^2 \frac{1}{(x-1)^2} \, dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).



13 Evaluate $\int_{-1}^{1} \frac{1}{x^{1/3}} \, dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.

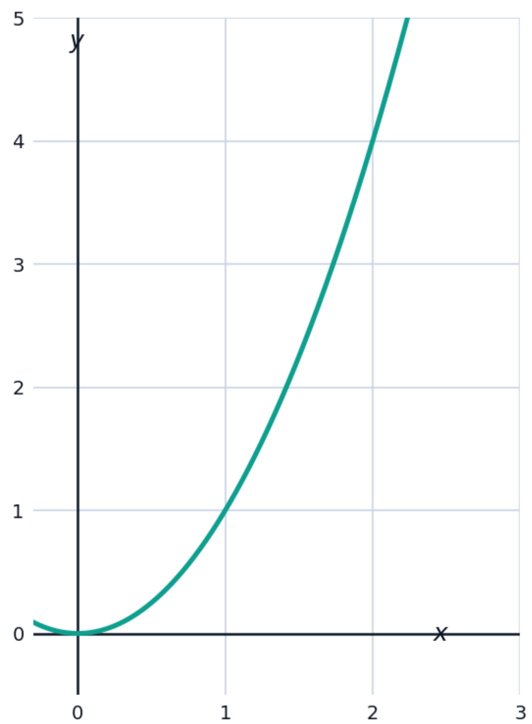
Visualizing an improper integral as a limiting area

14 The graph shows $f(x) = \frac{1}{x^2}$ on $[1, 10]$. As the right boundary extends to infinity, the shaded area approaches a finite limit. Explain why this is plausible, using the fact that $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (found earlier).



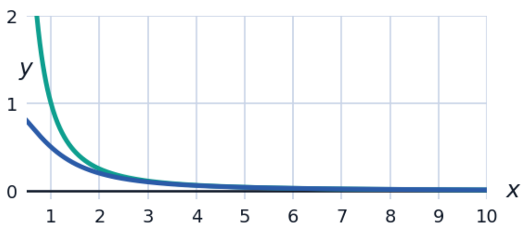
Arc length of a more involved curve

15 Set up (do not evaluate) the arc length integral for $y = x^2$ on $[0, 2]$, and estimate it using the straight-line distance between the endpoints as a lower bound.

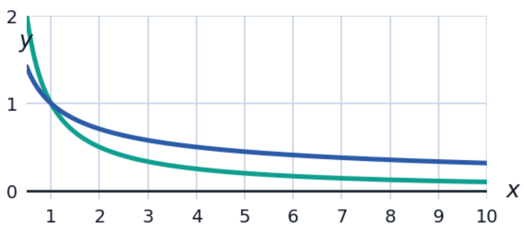


Comparison test for improper integrals

16 Use a comparison with $\int_1^\infty \frac{1}{x^2}\,dx$ (known to converge) to argue that $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges, without evaluating it.

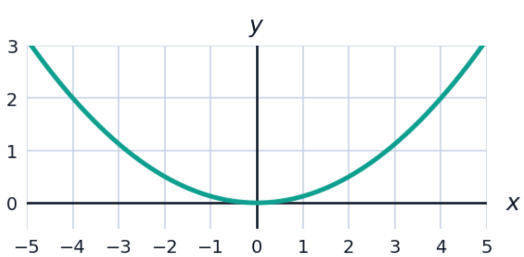


17 Use a comparison with $\int_1^\infty \frac{1}{x}\,dx$ (known to diverge) to argue that $\int_1^\infty \frac{1}{\sqrt{x}}\,dx$ also diverges.

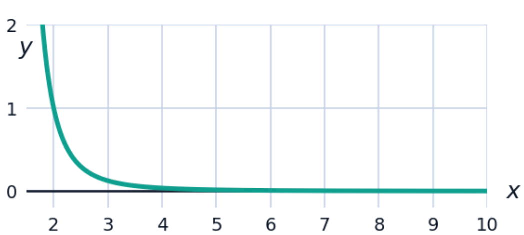


A capstone arc-length application

18 A cable hangs in the shape $y = \frac{x^2}{8}$ between the towers at $x = -4$ and $x = 4$. Set up the integral for the length of the cable (do not evaluate).



19 Evaluate $\int_2^\infty \frac{1}{(x-1)^3}\,dx$, or state that it diverges.



20 A city's water tower is shaped by revolving the curve $y = \sqrt{25 - x^2}$ (the upper half of a circle of radius 5) about the x-axis; set up (do not evaluate) the arc length integral for this curve on $[-4, 4]$, which describes the profile of one side of the tower.